

oI - Data & Knime

Alessandro Bregoli

2025

Introduction

In this tutorial we will **introduce** the **Knime** platform.



Introduction

In this tutorial we will **introduce** the **Knime** platform.

Specifically, the following arguments will be presented:

1. Knime Platform Installation & Introduction
- 2.
- 3.
- 4.
- 5.
- 6.



Introduction

In this tutorial we will **introduce** the **Knime platform**.

Specifically, the following arguments will be presented:

1. Knime Platform Installation & Introduction
2. Data Import & Exploration
- 3.
- 4.
- 5.
- 6.



Introduction

In this tutorial we will **introduce** the **Knime** platform.

Specifically, the following arguments will be presented:

1. Knime Platform Installation & Introduction
2. Data Import & Exploration
3. Missing Replacement
- 4.
- 5.
- 6.



Introduction

In this tutorial we will **introduce** the **Knime platform**.

Specifically, the following arguments will be presented:

1. Knime Platform Installation & Introduction
2. Data Import & Exploration
3. Missing Replacement
4. Sampling technique
- 5.
- 6.



Introduction

In this tutorial we will **introduce** the **Knime platform**.

Specifically, the following arguments will be presented:

1. Knime Platform Installation & Introduction
2. Data Import & Exploration
3. Missing Replacement
4. Sampling technique
5. Binarization & Discretization
- 6.



Introduction

In this tutorial we will **introduce** the **Knime platform**.

Specifically, the following arguments will be presented:

1. Knime Platform Installation & Introduction
2. Data Import & Exploration
3. Missing Replacement
4. Sampling technique
5. Binarization & Discretization
6. Variable Transformation



Introduction

In this tutorial we will **introduce** the **Knime platform**.

Specifically, the following arguments will be presented:

1. Knime Platform Installation & Introduction
2. Data Import & Exploration
3. Missing Replacement
4. Sampling technique
5. Binarization & Discretization
6. Variable Transformation

At the end of each section there will be a **hands on** to help you **familiarize** yourself with the **platform** and the **theoretical concepts**.



Knime Analytics Platform

Knime is a software that allows to *explore, analyze* and *exploit* data with a **low-code/no-code** approach.



Knime Analytics Platform

Knime is a software that allows to *explore, analyze* and *exploit* data with a **low-code/no-code** approach.

Knime provides a wide range of tools supporting the data scientist though all the data analysis steps such as:

- Data loading
- Data Exploration
- Model Estimation
- Preprocessing
- Visualization
- Model Validation



Knime Analytics Platform

Knime is a software that allows to *explore, analyze* and *exploit* data with a **low-code/no-code** approach.

Knime provides a wide range of tools supporting the data scientist though all the data analysis steps such as:

- Data loading
- Data Exploration
- Model Estimation
- Preprocessing
- Visualization
- Model Validation

Knime can be widely extended thanks to its plugin system that allows to import well known libraries and tools such as *Weka*.



Knime Analytics Platform

Knime is a software that allows to *explore, analyze* and *exploit* data with a **low-code/no-code** approach.

Knime provides a wide range of tools supporting the data scientist though all the data analysis steps such as:

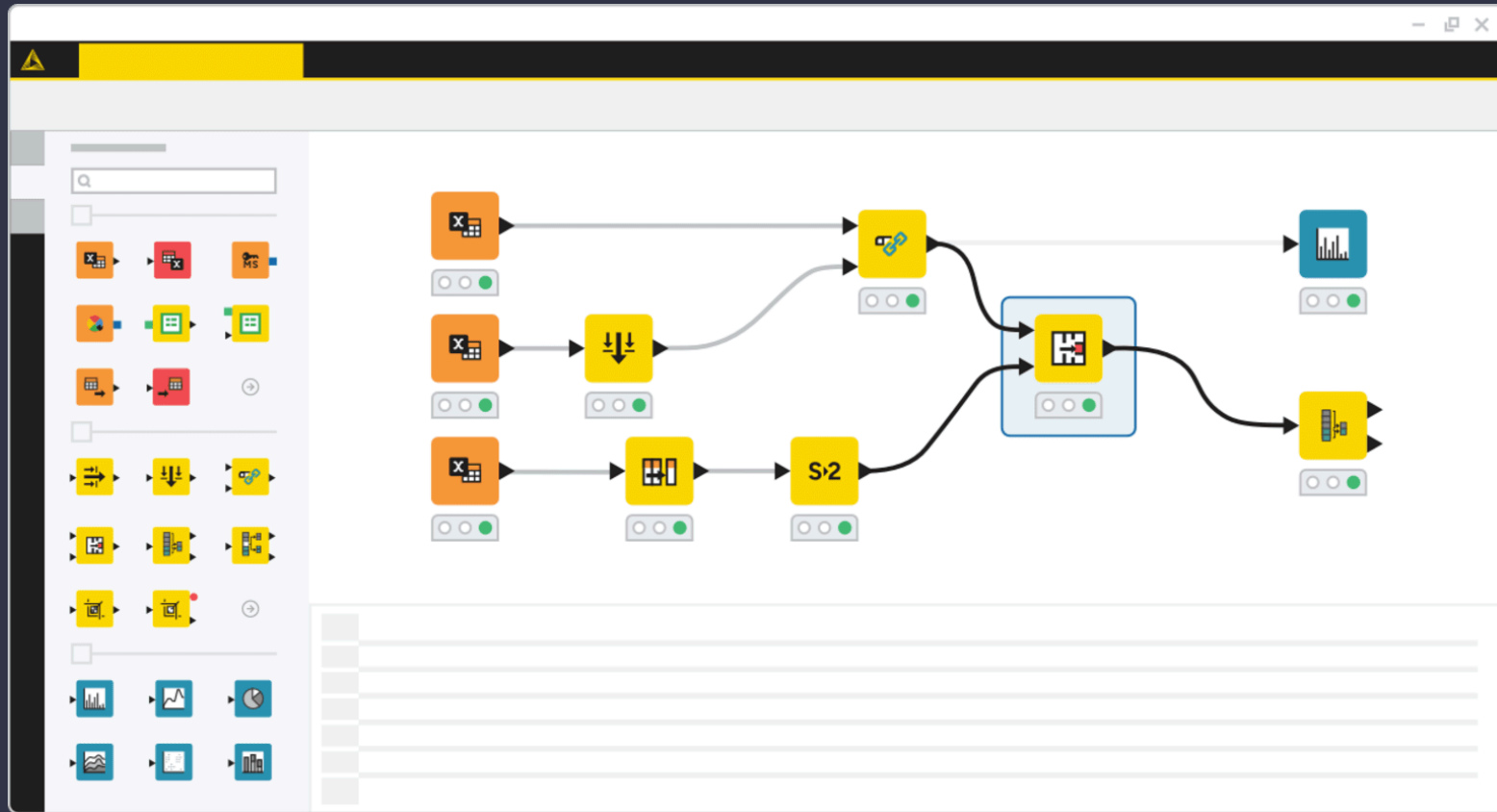
- Data loading
- Data Exploration
- Model Estimation
- Preprocessing
- Visualization
- Model Validation

Knime can be widely extended thanks to its plugin system that allows to import well known libraries and tools such as *Weka*.

Furthermore, the plugin system allows the use of programming languages such as: *Python* and *R*.



KNIME - workflow



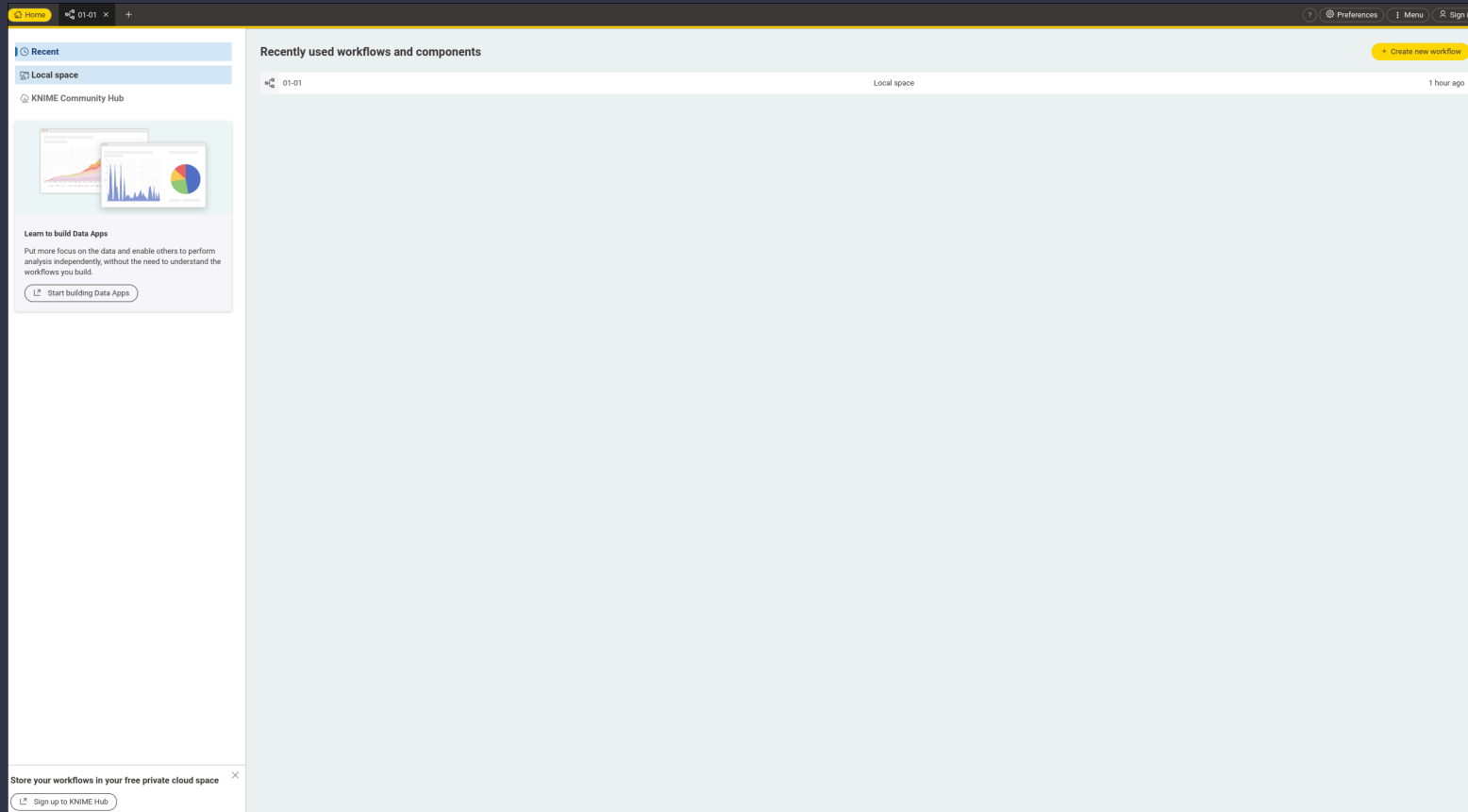
KNIME - Install



www.knime.com/downloads

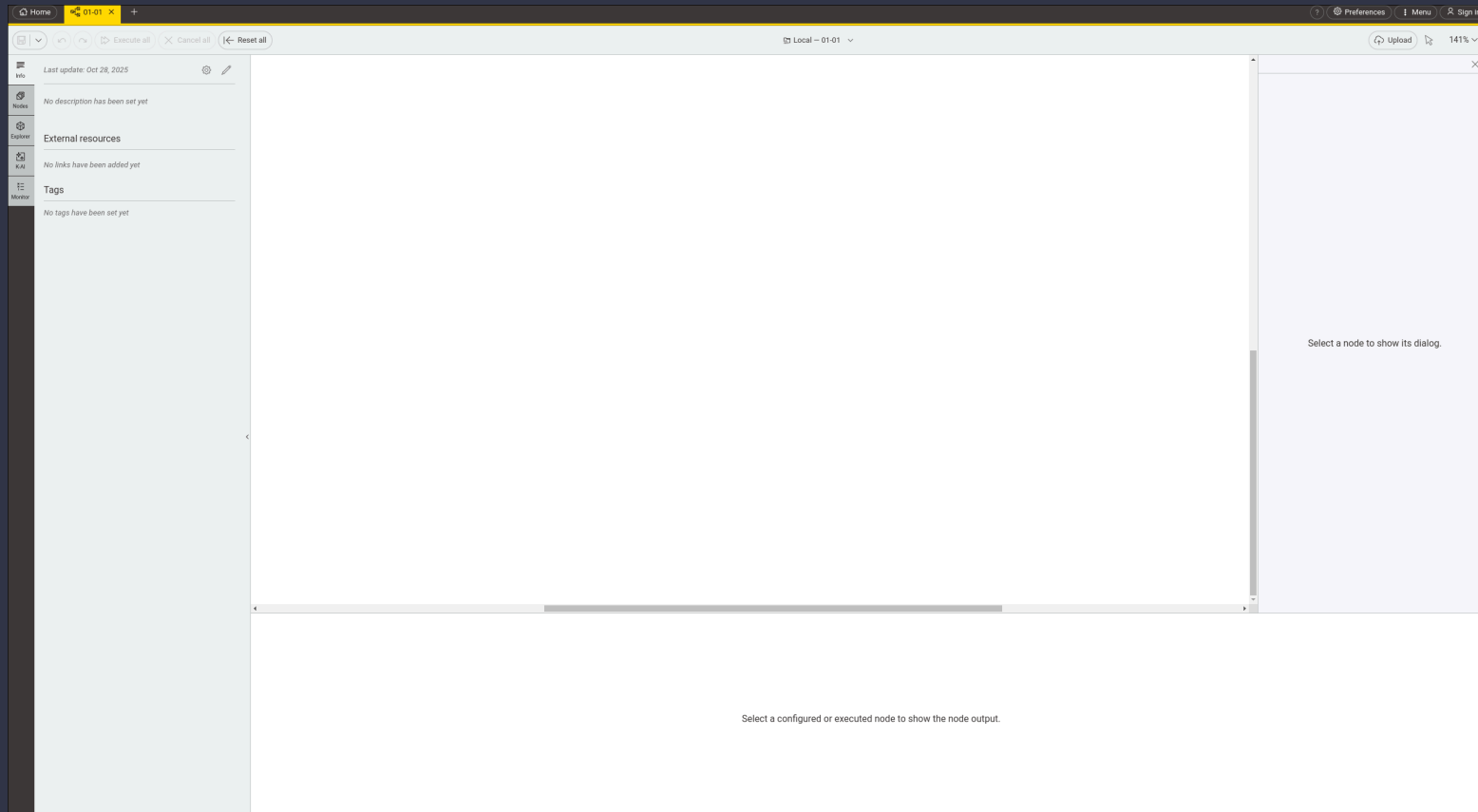
KNIME - Interface overview

Home



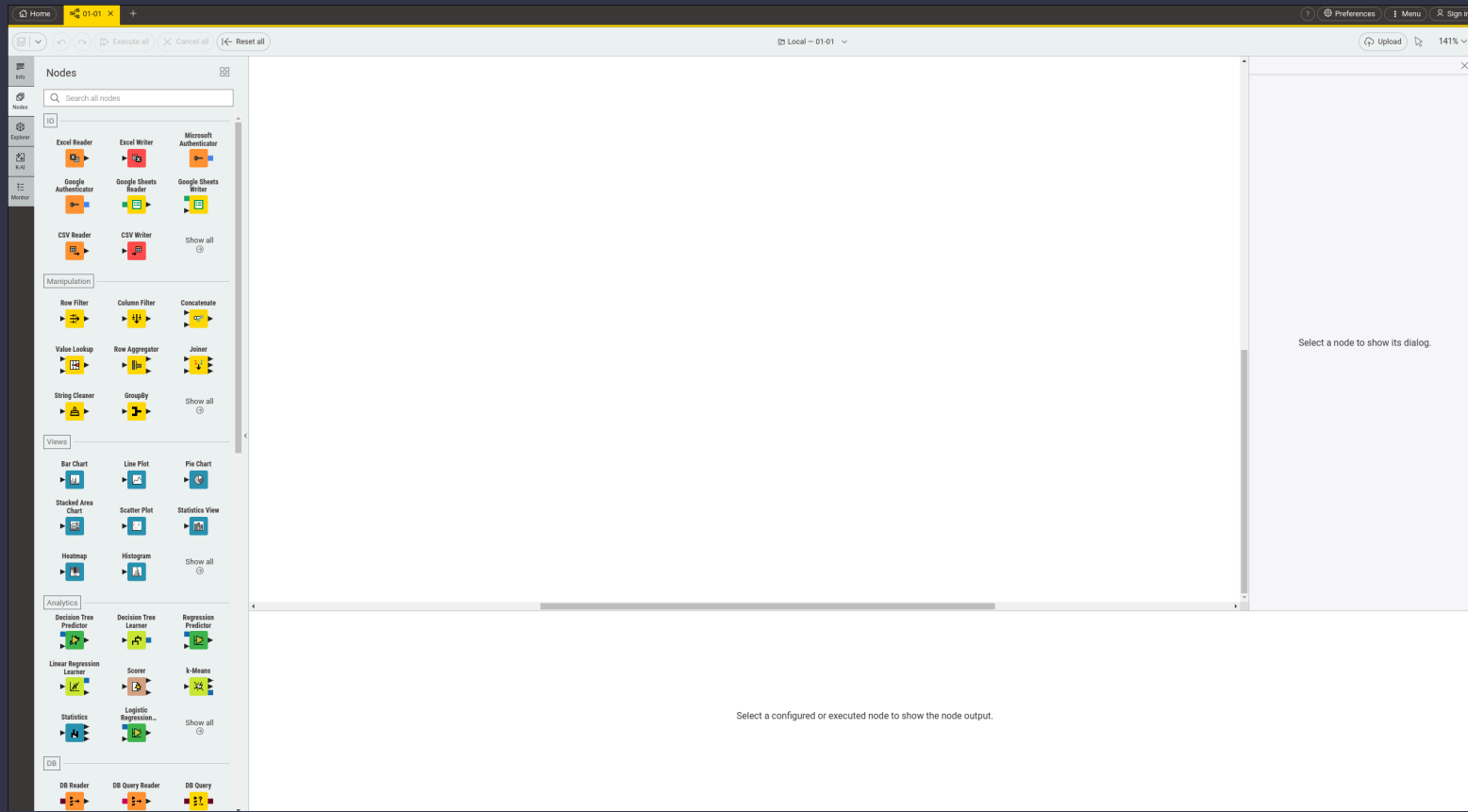
KNIME - Interface overview

Empty Workflow



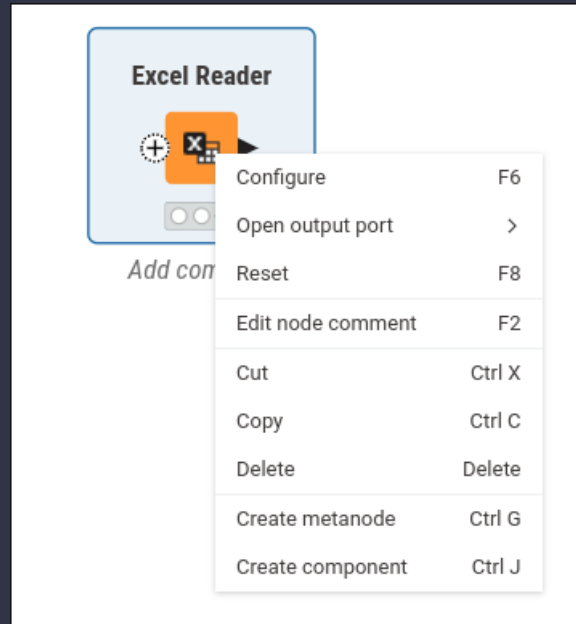
KNIME - Interface overview

Nodes



KNIME - Interface overview

Node option (right click)

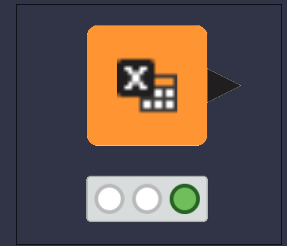


Data import & Exploration

Read data from file

The *Excel Reader* allows you to read the data from **xls/xlsx** files.

Excel Reader



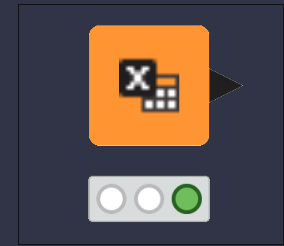
Read data from file

The *Excel Reader* allows you to read the data from **xls/xlsx** files.

The main configuration tab for this node are:

- File and Sheet
- Data Area
- Advanced
- Transformation

Excel Reader



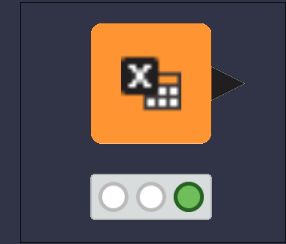
Read data from file

The *Excel Reader* allows you to read the data from **xls/xlsx** files.

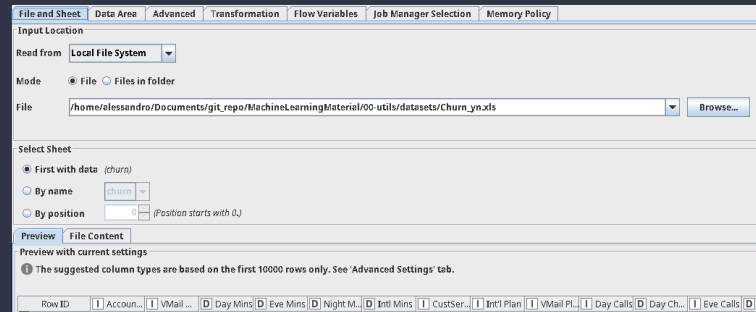
The main configuration tab for this node are:

- File and Sheet
- Data Area
- Advanced
- Transformation

Excel Reader



File and Sheet



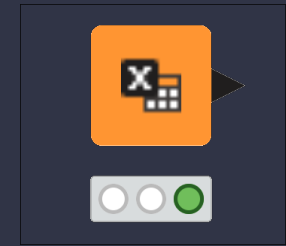
Read data from file

The *Excel Reader* allows you to read the data from **xls/xlsx** files.

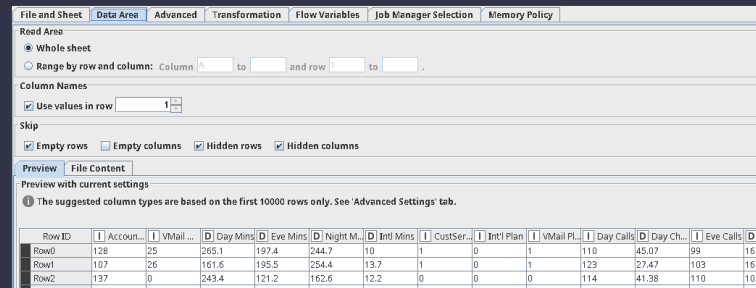
The main configuration tab for this node are:

- File and Sheet
- Data Area
- Advanced
- Transformation

Excel Reader



Data Area

The screenshot shows the 'Data Area' configuration tab of the Excel Reader node. It includes options for reading the whole sheet or a specific range, column names, and checkboxes for skipping empty rows, empty columns, hidden rows, and hidden columns. A preview table is shown at the bottom.

RowID	I Account...	I VMail ...	D Day Mins	D Eve Mins	D Night M...	D Intl Mins	I CustSer...	I Intl Plan	I VMail Pl...	I Day Calls	D Day Ch...	I Eve Calls	D E
Row0	128	25	265.1	197.4	244.7	10	1	0	1	110	45.07	99	16.7
Row1	107	26	161.6	195.5	254.4	13.7	1	0	1	123	27.47	103	16.6
Row2	137	0	243.4	121.2	162.6	12.2	0	0	0	114	41.38	110	10.3

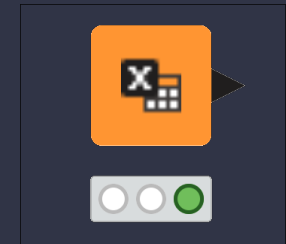
Read data from file

The *Excel Reader* allows you to read the data from **xls/xlsx** files.

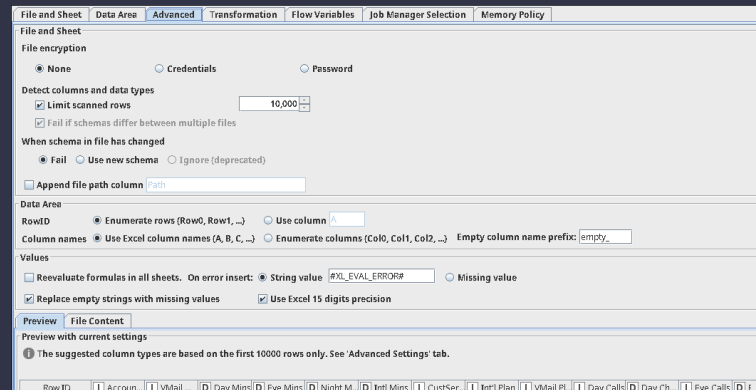
The main configuration tab for this node are:

- File and Sheet
- Data Area
- Advanced
- Transformation

Excel Reader



Advanced

The screenshot shows the 'Advanced' configuration tab of the Excel Reader node. It includes sections for 'File encryption' (None, Credentials, Password), 'Detect columns and data types' (Limit scanned rows: 10,000, Fail if schemas differ), 'When schema in file has changed' (Fail, Use new schema, Ignore), 'Data Area' (RowID, Column names, Values), and 'Preview' (File Content). The 'Preview' section shows a table with columns like Row ID, Account, VMail, Day Mins, Eve Mins, Night Mins, CustSer, Int1 Plan, VMail PL, Day Calls, Day Ch, and Eve Calls.

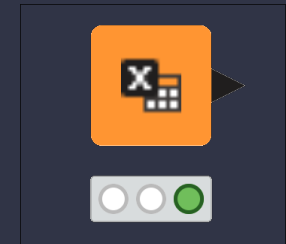
Read data from file

The *Excel Reader* allows you to read the data from **xls/xlsx** files.

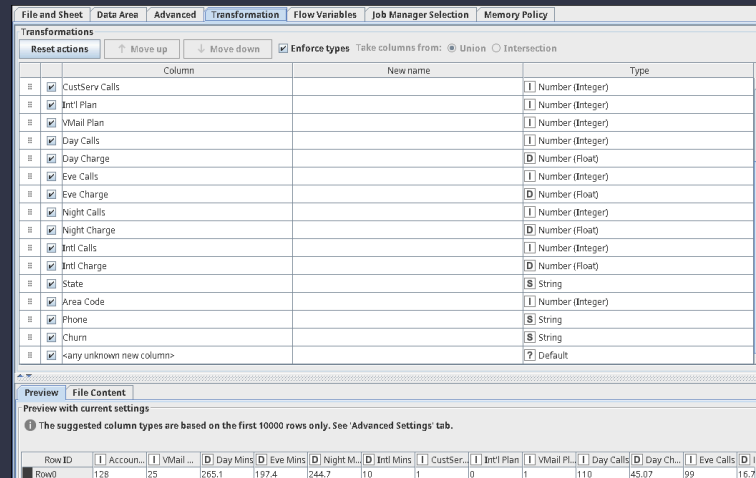
The main configuration tab for this node are:

- File and Sheet
- Data Area
- Advanced
- Transformation

Excel Reader



Transformation

The screenshot shows the 'Transformation' tab of the Excel Reader node configuration. It includes a table for mapping columns to new names and types, and a preview section at the bottom showing a sample of the data being read.

Column	New name	Type
☒ CustServ Calls		☐ Number (Integer)
☒ Intl Plan		☐ Number (Integer)
☒ VMail Plan		☐ Number (Integer)
☒ Day Calls		☐ Number (Integer)
☒ Day Charge		☐ Number (Float)
☒ Eve Calls		☐ Number (Integer)
☒ Eve Charge		☐ Number (Float)
☒ Night Calls		☐ Number (Integer)
☒ Night Charge		☐ Number (Float)
☒ Intl Calls		☐ Number (Integer)
☒ Intl Charge		☐ Number (Float)
☒ State		☐ String
☒ Area Code		☐ Number (Integer)
☒ Phone		☐ String
☒ Churn		☐ String
☒ -any unknown new column-		☐ Default

Row ID	Account...	VMail...	Day Mins	Eve Mins	Night M...	Intl Mins	CustSer...	Intl Plan	VMail PL...	Day Calls	Day Ch...	Eve Calls	I
Row0	128	25	265.1	197.4	244.7	10	1	0	1	110	45.07	99	16.7

Summary Statistics

The *Statistic* node, allows to extract from the connected table a set of statistics:

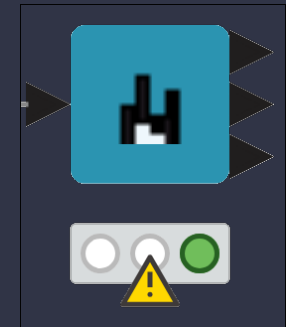
Numeric Variables

- Mean
- Max
- Min
- Quartiles
- Histogram

Nominal Variables

- Number of missing
- Histogram

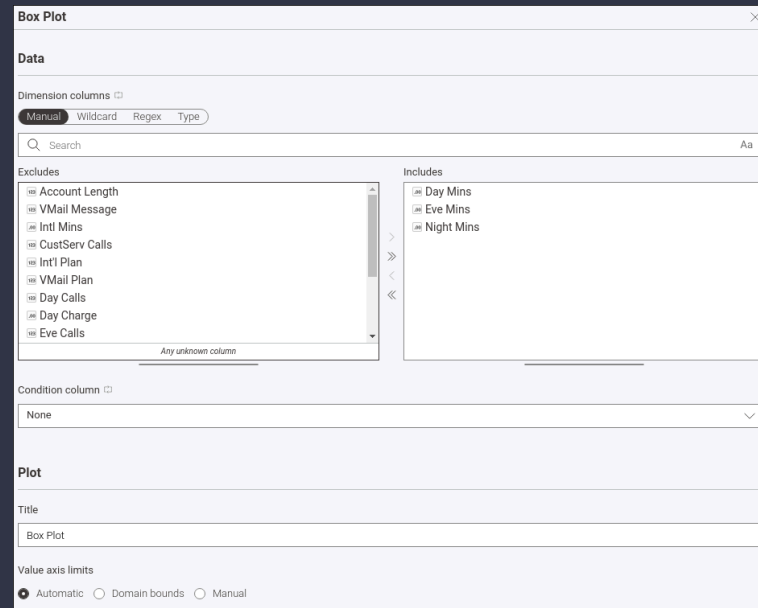
Statistics



Visualization - Box plot

The *Boxplot* node allows to select a subset of variables and graphically shows their distribution with a boxplot.

Config



The screenshot shows the 'Box Plot' configuration window. It is divided into two main sections: 'Data' and 'Plot'.

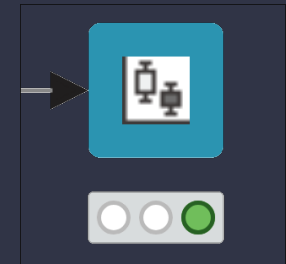
Data Section:

- Dimension columns:** A search bar with 'Search' and 'Aa' buttons. Below it are tabs for 'Manual', 'Wildcard', 'Regex', and 'Type'. The 'Manual' tab is selected.
- Excludes:** A list of columns to be excluded from the plot. The list includes: Account Length, VMail Message, Int'l Mins, CustServ Calls, Int'l Plan, VMail Plan, Day Calls, Day Charge, and Eve Calls. A scroll bar is on the right.
- Includes:** A list of columns to be included in the plot. The list includes: Day Mins, Eve Mins, and Night Mins.
- Condition column:** A dropdown menu currently set to 'None'.

Plot Section:

- Title:** A text field containing 'Box Plot'.
- Value axis limits:** Three radio buttons: 'Automatic' (selected), 'Domain bounds', and 'Manual'.

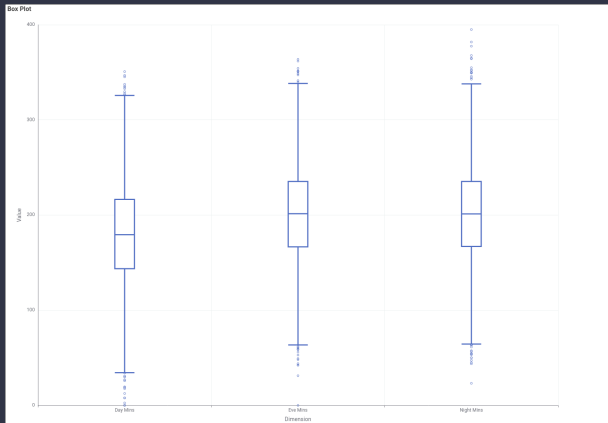
Box plot



Visualization - Box plot

The *Boxplot* node allows to select a subset of variables and graphically shows their distribution with a boxplot.

Plot



Box

Spans from Q1 and Q3

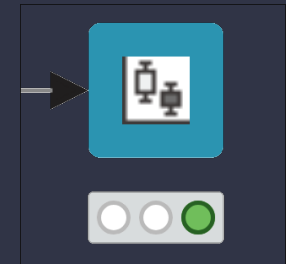
Central Line

Median

Whiskers

$-1.5 \cdot IQR$ to $1.5 \cdot IQR$

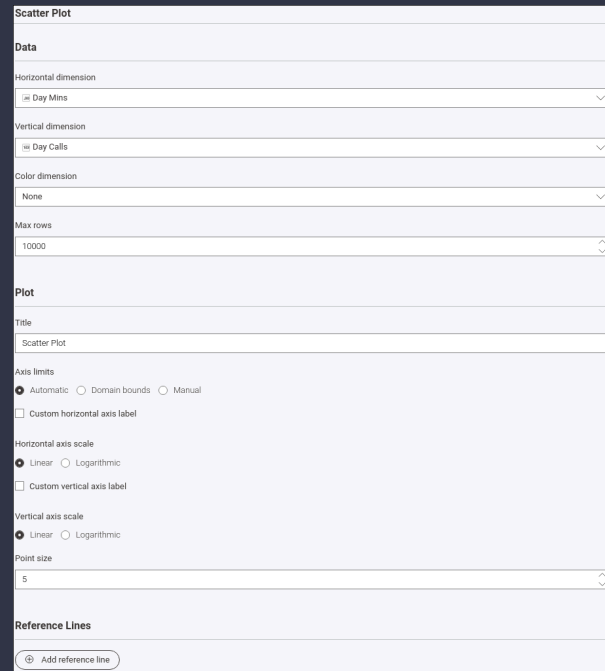
Box plot



Visualization - Scatter plot

The *scatter plot* node is a visual tool to quickly access the relation between two variables.

Config



The image shows the configuration window for the 'Scatter Plot' node. It is divided into several sections: 'Data', 'Plot', and 'Reference Lines'. The 'Data' section includes dropdowns for 'Horizontal dimension' (set to 'Day Mins'), 'Vertical dimension' (set to 'Day Calls'), and 'Color dimension' (set to 'None'). It also has a 'Max rows' dropdown set to '10000'. The 'Plot' section includes a 'Title' field (set to 'Scatter Plot'), 'Axis limits' (set to 'Automatic'), 'Horizontal axis scale' (set to 'Linear'), and 'Vertical axis scale' (set to 'Linear'). There are checkboxes for 'Custom horizontal axis label' and 'Custom vertical axis label', both of which are unchecked. The 'Point size' is set to '5'. The 'Reference Lines' section has a button labeled 'Add reference line'.

Scatter Plot

Data

Horizontal dimension
Day Mins

Vertical dimension
Day Calls

Color dimension
None

Max rows
10000

Plot

Title
Scatter Plot

Axis limits
☒ Automatic ☐ Domain bounds ☐ Manual

☐ Custom horizontal axis label

Horizontal axis scale
☒ Linear ☐ Logarithmic

☐ Custom vertical axis label

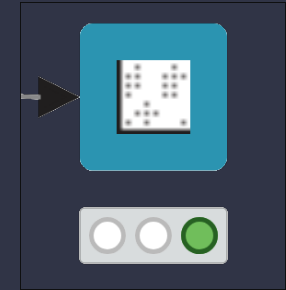
Vertical axis scale
☒ Linear ☐ Logarithmic

Point size
5

Reference Lines

Add reference line

Scatter plot



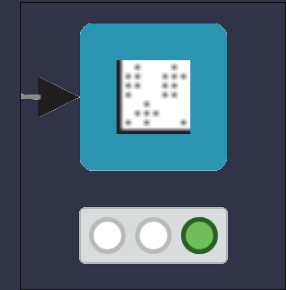
Visualization - Scatter plot

The *scatter plot* node is a visual tool to quickly access the relation between two variables.

View



Scatter plot



Hands on (15 minutes)

- Download the **churn_yn.xls** dataset from **elearning**.
- Import the file using the **Excel reader**
- Explore the content of the file with:
 - Statistics
 - Boxplot
 - Scatterplot



Missing Replacement

Introduction

The presence of **missing data** in a dataset is a quite common situation that can arise from many causes such as:

- Data collection problem
- Evolving data collection process
- Inconsistent values



Introduction

The presence of **missing data** in a dataset is a quite common situation that can arise from many causes such as:

- Data collection problem
- Evolving data collection process
- Inconsistent values

Missing data can be classified as follows:

- **Missing Completely At Random (MCAR)**
- Missing At Random (MAR)
- Missing Not At Random (MNAR)



Introduction

The presence of **missing data** in a dataset is a quite common situation that can arise from many causes such as:

- Data collection problem
- Evolving data collection process
- Inconsistent values

Missing data can be classified as follows:

- **Missing Completely At Random (MCAR)**
- Missing At Random (MAR)
- Missing Not At Random (MNAR)

The **missing data** problem must be **addressed** at the **beginning** of any **data mining study**.



Missing Value Node

The *missing value* node is the most simple instrument provided by knime to deal with missing data.

Missing Value



Missing Value Node

The *missing value* node is the most simple instrument provided by knime to deal with missing data.

The missing data handler provided by this node are:

- Remove Row
- Fix Value
- Next
- Previous
- Mean
- Median
- Maximum
- Minimum
- Average interpolation
- Most Frequent Value
- Moving Average

Missing Value



Missing Value Node

The *missing value* node is the most simple instrument provided by knime to deal with missing data.

The missing data handler provided by this node are:

- Remove Row
- Fix Value
- Next
- Previous
- Mean
- Median
- Maximum
- Minimum
- Average interpolation
- Most Frequent Value
- Moving Average

These handler can be applied in two ways:

- Over columns with a specific type (*Integer, Double, String*)
- Over specific columns

Missing Value



Conditional Replacement

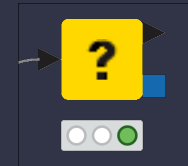
The *conditional replacement* approach consist in **partitioning** the dataset and **deal** with **missing** value **independently** in each **partition**.

- **Row filter**: This node allows to select subset of the dataset **filtering** out the **rows** that does not match the **condition** of interest.
- **Concatenate**: concatenate the rows of two tables

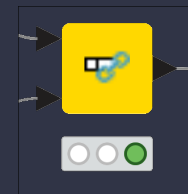
Row Filter



Missing Value



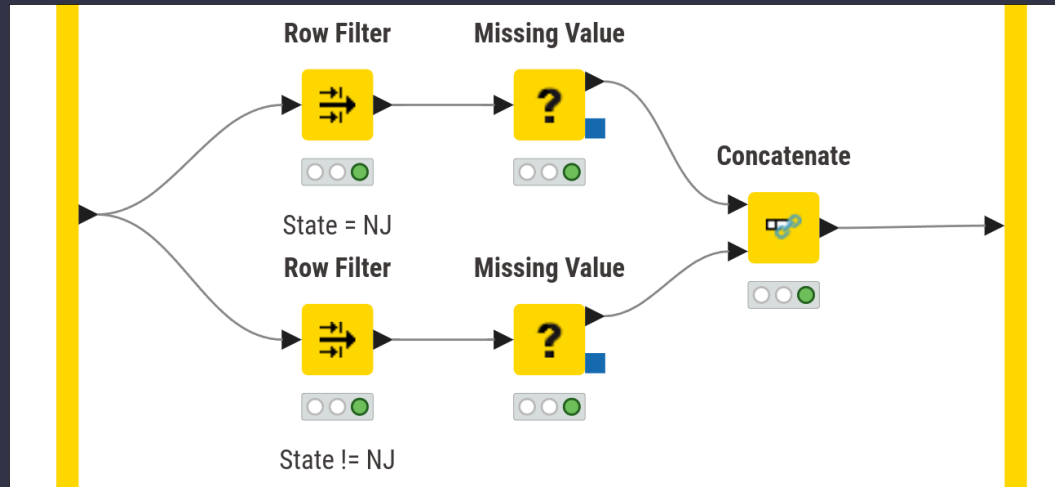
Concatenate



Conditional Replacement

The *conditional replacement* approach consist in **partitioning** the dataset and **deal** with **missing value independently in each partition**.

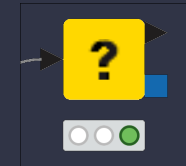
- **Row filter**: This node allows to select subset of the dataset **filtering** out the **rows** that does not match the **condition** of interest.
- **Concatenate**: concatenate the rows of two tables



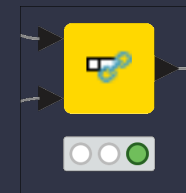
Row Filter



Missing Value



Concatenate



Most Probable

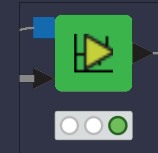
The *most probable* method works by focusing on the **relationship between two variables**: we use the value of the known variable to predict the value of the missing one.

- **Linear Regression Learner**: fit a linear regression
- **Regressor Predictor**: estimate the value of the dependent variable for a regression model.
- **Column Filter**: remove columns based on a user defined condition
- **Column Renamer**: Rename a column

Linear Regression Learner



Regressor Predictor



Column Filter



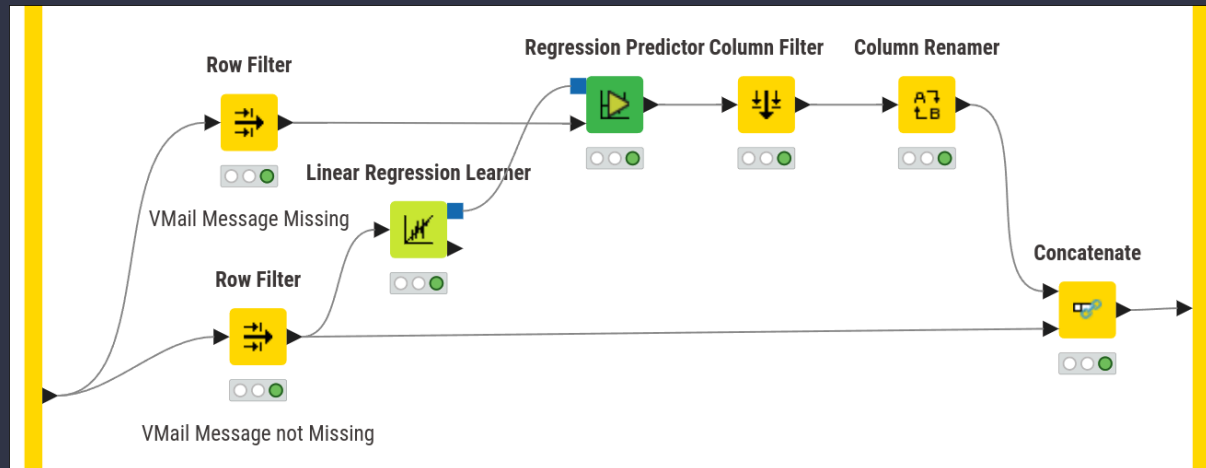
Column Renamer



Most Probable

The *most probable* method works by focusing on the **relationship between two variables**: we use the value of the known variable to predict the value of the missing one.

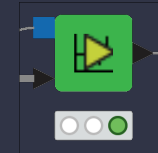
Predicting the missing values for the variable **VMail Message** using the value of the variable **VMail plan**.



Linear Regression Learner



Regressor Predictor



Column Filter



Column Renamer



Hands on (30 minutes)

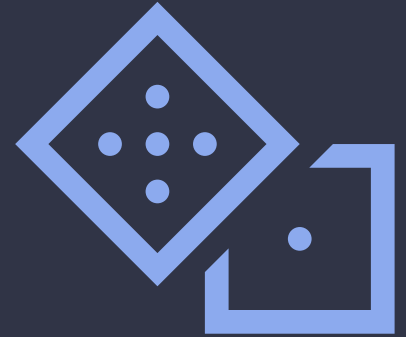
- Download the **Churn_yn_missing_data.csv** dataset from **elearning**.
- **Import** the file using the **CSV reader**
- **Apply** the following methods for dealing with **missing values**:
 - Remove Row
 - *hint: use the **missing value** node.*
 - Conditional Replacement
 - *hint: use two **filter row** nodes, two **missing value** nodes and a **concatenate** node.*
 - Most Probable Replacement
 - *hint: the structure is similar to the previous one but you have to use **Linear Regression Learner** node and a **Regressor Predictor** node.*



Preprocessing - Sampling

Introduction

The sampling process consists in selecting a set of samples (rows) from a population.

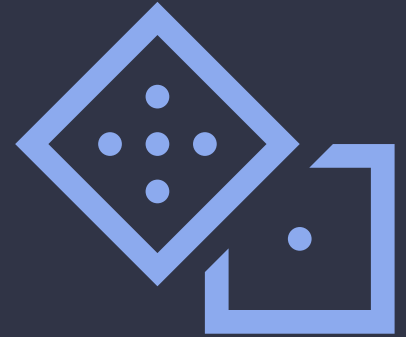


Introduction

The sampling process consists in selecting a set of samples (rows) from a population.

The motivations for applying a sampling technique can be for example:

- Quick analysis over a huge dataset
- Increase the size of a small dataset
- Validate a model over different portion of the dataset



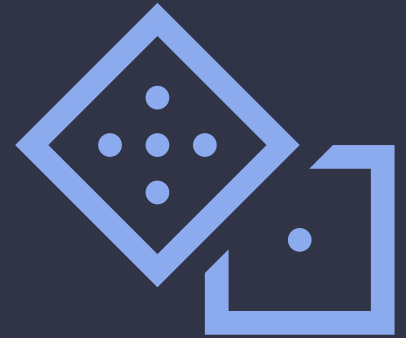
Introduction

The sampling process consists in selecting a set of samples (rows) from a population.

The motivations for applying a sampling technique can be for example:

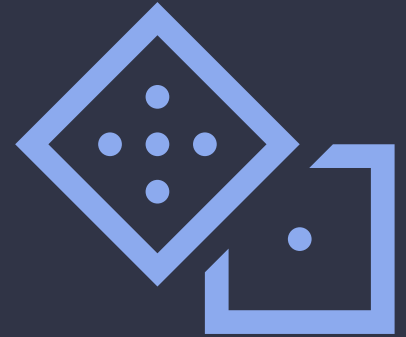
- Quick analysis over a huge dataset
- Increase the size of a small dataset
- Validate a model over different portion of the dataset

Regardless of the reason behind the sampling, usually the user is interested in generating a sample that is **representative** for the entire dataset.



Sampling & Pseudo Random Numbers

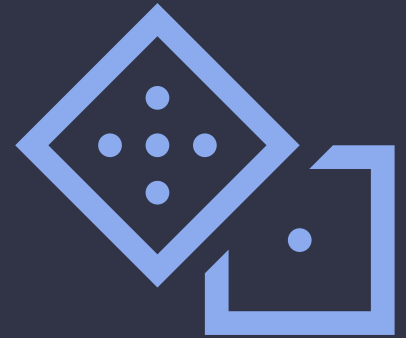
Many of the sampling methods presented in this section require randomness.



Sampling & Pseudo Random Numbers

Many of the sampling methods presented in this section require randomness.

However, computer are only able to generate **pseudo-random sequences**; sequence of numbers number that approximate the behavior of a sequence of real random numbers.

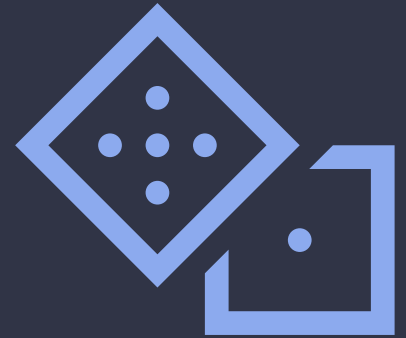


Sampling & Pseudo Random Numbers

Many of the sampling methods presented in this section require randomness.

However, computer are only able to generate **pseudo-random sequences**; sequence of numbers number that approximate the behavior of a sequence of real random numbers.

This means that, if we fix the **seed** the sequence of random number is deterministic.



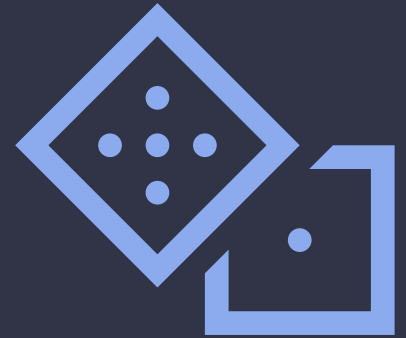
Sampling & Pseudo Random Numbers

Many of the sampling methods presented in this section require randomness.

However, computer are only able to generate **pseudo-random sequences**; sequence of numbers number that approximate the behavior of a sequence of real random numbers.

This means that, if we fix the **seed** the sequence of random number is deterministic.

In general this can be a limitation. However, in our case, setting the seed allows us to achieve **replicability**.



Sampling & Pseudo Random Numbers

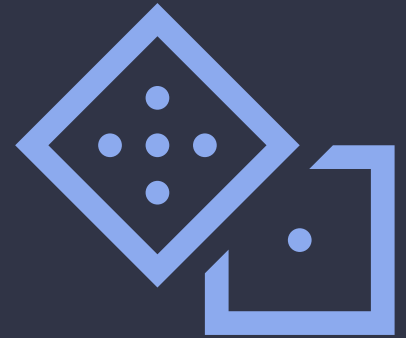
Many of the sampling methods presented in this section require randomness.

However, computer are only able to generate **pseudo-random sequences**; sequence of numbers number that approximate the behavior of a sequence of real random numbers.

This means that, if we fix the **seed** the sequence of random number is deterministic.

In general this can be a limitation. However, in our case, setting the seed allows us to achieve **replicability**.

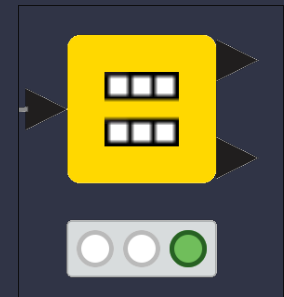
All the **knime nodes** relying on **randomness** permit to set the **seed**.



Partitioning

The *Table Partitioner* node, allows to split a dataset in two distinct partitions.

Table Partitioner



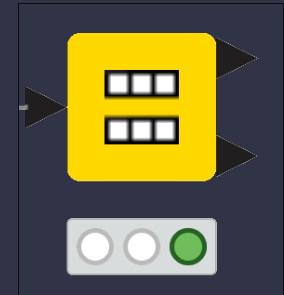
Partitioning

The *Table Partitioner* node, allows to split a dataset in two distinct partitions.

The user can define both the size of the first partition and one of the following sampling methods:

- **First rows:** Allows you to select the first top rows of the input table.
- **Random:** Randomly selects rows.
- **Stratified:** Preserves the distribution of values in the selected group column.
- **Linear:** Selects rows evenly spaced across the input table, always including the first and last row.

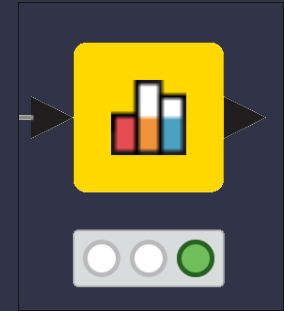
Table Partitioner



Equal Size Sampling

The *Equal Size Sampling* node implement a particular kind of **undersampling**. Specifically, it removes rows from the input data set such that the values in a categorical column are equally distributed.

Equal Size Sampling

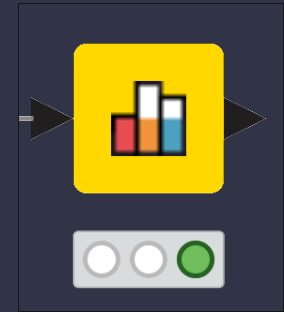


Equal Size Sampling

The *Equal Size Sampling* node implement a particular kind of **undersampling**. Specifically, it removes rows from the input data set such that the values in a categorical column are equally distributed.

This approach **does not generate** a **representative** sample for the original dataset.

Equal Size Sampling



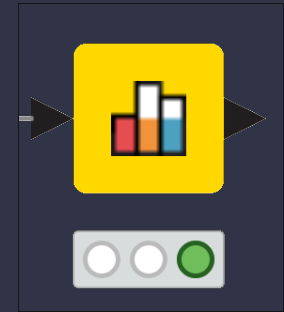
Equal Size Sampling

The *Equal Size Sampling* node implement a particular kind of **undersampling**. Specifically, it removes rows from the input data set such that the values in a categorical column are equally distributed.

This approach **does not generate** a **representative** sample for the original dataset.

In the next lectures we will see that this tool can be useful to learn models where the **class of interest** is highly **under-represented**.

Equal Size Sampling



Bootstrap Sampling

The *Bootstrap Sampling* node randomly draws rows from the input with replacement. The output will therefore likely contain duplicate rows while other rows are not present in the output at all.

Bootstrap Sampling



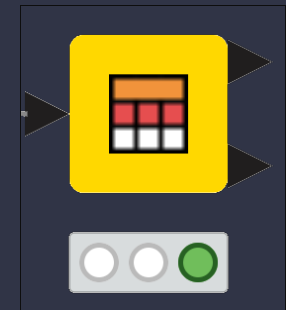
Bootstrap Sampling

The *Bootstrap Sampling* node randomly draws rows from the input with replacement. The output will therefore likely contain duplicate rows while other rows are not present in the output at all.

The node generate two partitions:

- The **first** partition contains the **rows sampled with repetition** from the original dataset.
- The second partition contains the **rows that never appear in the first one**. This table **can be empty**.

Bootstrap Sampling



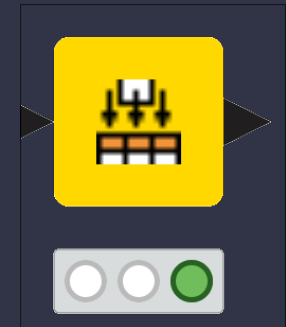
Preprocessing - Binarization & Discretization

One Hot Encoding

There are situations in which the possible values assumed by a categorical variable cannot be ordered:

Color	ID	Red	Blue	Green
Red	0	1	0	0
Blue	1	0	1	0
Green	2	0	0	1

One to Many



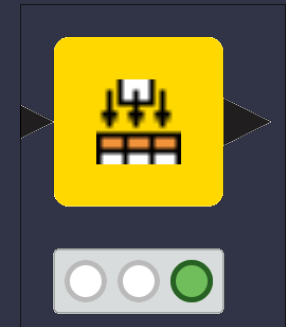
One Hot Encoding

There are situations in which the possible values assumed by a categorical variable cannot be ordered:

Color	ID	Red	Blue	Green
Red	0	1	0	0
Blue	1	0	1	0
Green	2	0	0	1

The *One to Many* node allows us to select any **String** column and generate a column for each possible value that column can assume.

One to Many

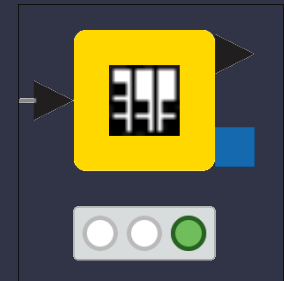


Discretization

Discretization can be useful in many situations:

- Improve performance of a machine learning algorithm
- Deal with outliers
- Increase interpretability
- Exploit expert knowledge

Binner



Discretization

Discretization can be useful in many situations:

- Improve performance of a machine learning algorithm
- Deal with outliers
- Increase interpretability
- Exploit expert knowledge

The *Binner* allows to discretize a numeric column with the following strategies:

- **Custom Cutoffs:** Manual discretization.
- **Equal Size:** Each bin has the same width.
- **Equal Frequency:** Each bin represent the same number of rows.
- **Custom Quantiles:** Each bin represent a manually specified percentage of rows.

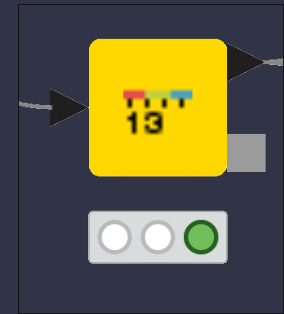
Binner



Supervised binning

The *CAIM Binner* node implements a particular kind of **supervised discretization**.

Caim Binner



Supervised binning

The *CAIM Binner* node implements a particular kind of **supervised discretization**.

A supervised discretization algorithm exploit additional information from a categorical (class) distribution.

Caim Binner



Supervised binning

The *CAIM Binner* node implements a particular kind of **supervised discretization**.

A supervised discretization algorithm exploit additional information from a categorical (class) distribution.

Specifically, the algorithm create bins where the instances within a bin predominantly belong to one class.

Caim Binner



Hands on (30 minutes)

- Download the **Churn_yn.xls** dataset from **elearning**.
- Apply the following discretization methods to the **Day Mins** variable:
 - Equal Size
 - Equal Frequency
 - CAIM Binner
 - *hint: use churn as class*
- Evaluate the distribution of the discretized variable on **churn** and **non churn** customers
 - *hint: use a row filter node in combination with a Pie Chart node*

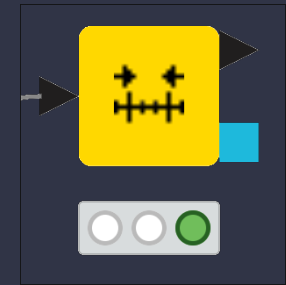


Preprocessing - Variable Transformation

Normalizer

The *Normalizer* node implements two common transformations for continuous variables:

Normalizer



Normalizer

The *Normalizer* node implements two common transformation for continuous variables:

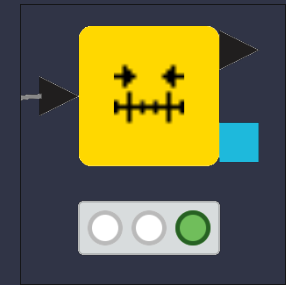
Normalization:

$$Z = \frac{x - \hat{\mu}}{\hat{\sigma}}$$

Min-Max Scaler:

$$M = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Normalizer



Normalizer

The *Normalizer* node implements two common transformation for continuous variables:

Normalization:

$$Z = \frac{x - \hat{\mu}}{\hat{\sigma}}$$

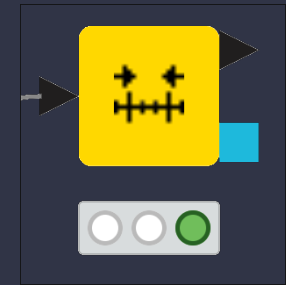
Min-Max Scaler:

$$M = \frac{x - \min(x)}{\max(x) - \min(x)}$$

The **Normalization**, transforms the values to have the following properties:

- **Mean** equal to 0
- **Standard Deviation** equal to 1

Normalizer



Normalizer

The *Normalizer* node implements two common transformation for continuous variables:

Normalization:

$$Z = \frac{x - \hat{\mu}}{\hat{\sigma}}$$

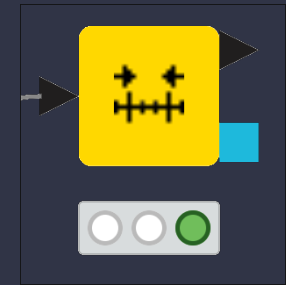
Min-Max Scaler:

$$M = \frac{x - \min(x)}{\max(x) - \min(x)}$$

The **Min-Max Scaler**: adjust the values such that it has the following properties:

- **Min**: 0
- **Max**: 1

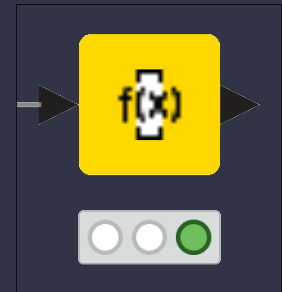
Normalizer



Expression

The *Expression* node enable generic row by row manipulation.

Expression

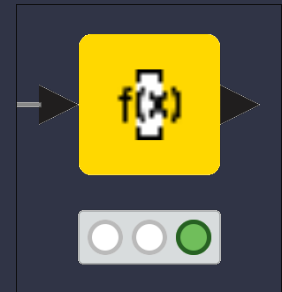


Expression

The *Expression* node enable generic row by row manipulation.

All the most common mathematical functions and constants are supported.

Expression



Expression

The *Expression* node enable generic row by row manipulation.

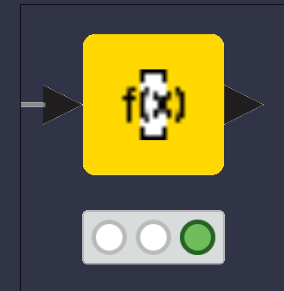
All the most common mathematical functions and constants are supported.

Expression Config

The screenshot shows the 'Expression' node configuration in Knime. It features a left sidebar with a tree view of input tables and flow variables. The main area contains two expression editors. 'Expression 1' contains a multi-line comment and a formula: `$["Day Mins"]/$["Day Calls"]`. Below it, the output is configured to create a new column named 'Day_Call_Length'. 'Expression 2' contains the formula: `($["Day Mins"] - COLUMN_MIN("Day Mins"))/(COLUMN_MAX("Day Mins") - COLUMN_MIN("Day Mins"))`. A right sidebar provides a search bar and a list of mathematical functions under the 'Math - General' category, including pow, sqrt, mod, abs, sign, exp, ln, log10, log2, and log. Below this, the 'COLUMN_MIN' function is detailed. At the bottom, there is a preview table showing data for 10 rows, with columns for various metrics like Account, VMail, Day Mins, Eve Mins, Night Mins, Intl Mins, CustServ, Intl Plan, VMail Plan, Day Calls, Day Char, Eve Calls, and Eve Mins. Buttons for 'Ask K-AI', 'Evaluate first 10 rows', 'Apply and Execute', and 'Apply' are visible.

RowID	Account ...	VMail Me...	Day Mins	Eve Mins	Night Mins	Intl Mins	CustServ ...	Intl Plan	VMail Plan	Day Calls	Day Char...	Eve Calls	Eve Mins
Row0	128	25	265.1	197.4	244.7	10	1	0	1	110	45.07	99	14
Row1	107	26	161.6	195.5	254.4	13.7	1	0	1	123	27.47	103	14
Row2	137	0	243.4	121.2	162.6	12.2	0	0	0	114	41.38	110	14
Row3	84	0	299.4	61.9	196.9	6.6	2	1	0	71	50.9	88	5
Row4	75	0	166.7	148.3	186.9	10.1	3	1	0	113	28.34	122	1
Row5	118	0	223.4	220.6	203.9	6.3	0	1	0	98	37.98	101	11
Row6	121	24	218.2	348.5	212.6	7.5	3	0	1	88	37.09	108	21

Expression



Hands on (30 minutes)

1. Download the **Churn_ys.xls** from elearning.
2. Use the **Expresion** node to replicate the following transformations on the **VMail Message** variable:
 - Normalization: $Z = \frac{x - \hat{\mu}}{\hat{\sigma}}$
 - Min-Max Scaler: $M = \frac{x - \min(x)}{\max(x) - \min(x)}$
3. Use the **Statistics** node to evaluate property of the normalized and scaled columns.
4. Use the **Expression** node to compute the following quantities:
 - Mean duration of a Day call for each customer.
 - Mean duration of a Eve call for each customer.
 - Mean duration of a Night call for each customer.
5. Use a **Box plot** node to shows the distribution of the new variables using **Churn** as **Condition Column**.



Thanks!